Research Methodology

Answers to the 5 Most Important Questions from Each Unit

MDS 601 - Exam-Oriented Answer Guide

This guide gives concise but complete answers to the selected high-priority questions. The wording is kept close to examination style: definition first, then points, examples, and conclusion where needed.

Unit 1: Introduction to Research Methodology

1.1 What is scientific research? Describe the major steps involved in the scientific research process.

Scientific research is a systematic, objective and logical process of inquiry used to discover facts, verify existing knowledge, solve problems and develop or refine theories. It follows planned procedures rather than guesswork. The researcher begins with a problem, gathers evidence, analyzes the evidence and reaches a conclusion that can be checked by others.

Scientific Research Process

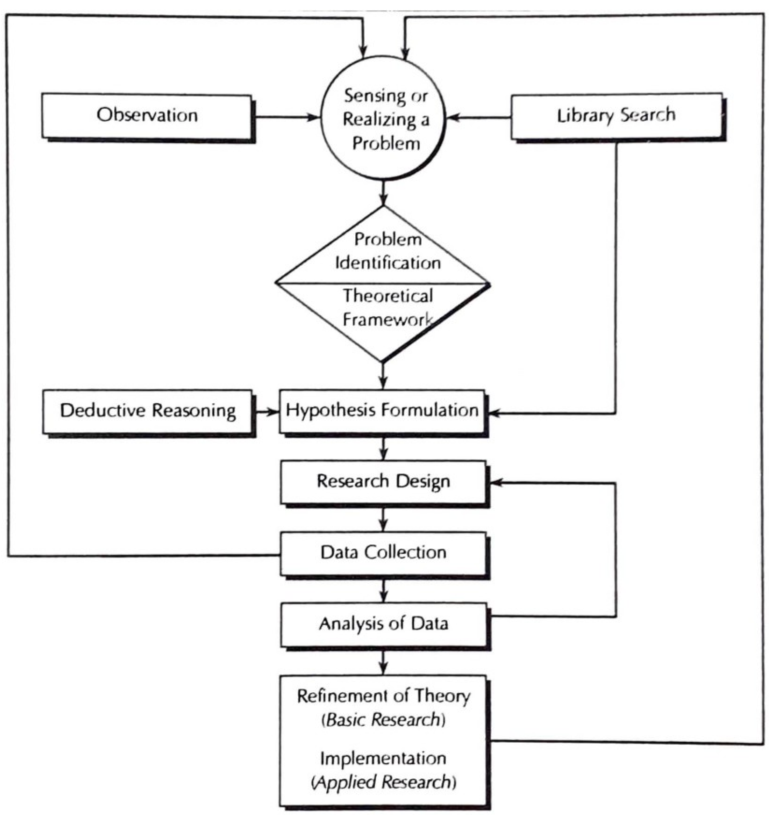

Figure: Scientific research process (from the study material, page 7).

Major steps in the scientific research process are:

1. **Sensing or realizing a problem:** The researcher observes a situation and notices an issue that needs investigation.
2. **Problem identification:** The broad issue is narrowed into a clear and researchable problem. A well-defined problem gives direction to the whole study.

3. **Theoretical framework:** Relevant concepts, variables and possible relationships are arranged logically to explain the problem.
4. **Hypothesis formulation:** A tentative, testable statement about the relationship between variables is developed.
5. **Research design:** The researcher prepares the plan for data collection, measurement, sampling and analysis.
6. **Data collection:** Required primary or secondary data are collected using suitable tools such as questionnaires, interviews, observation or records.
7. **Data analysis:** Data are edited, coded, tabulated and analyzed using appropriate statistical or logical methods.
8. **Refinement of theory or practice:** The findings are interpreted. In basic research, theory may be refined; in applied research, practical recommendations may be made.

   Exam point: Scientific research is systematic, cyclical and self-correcting. It starts with a problem and ends with findings that may create another research problem.

1.2 Explain the characteristics of scientific method.

Scientific method is the organized procedure used to acquire reliable knowledge. Its major characteristics are:

- **Verifiability:** Findings must be capable of being checked again by other researchers. The phenomenon should be observable or measurable directly or indirectly.
- **Generality:** Scientific findings should have wider applicability beyond a single individual case, although in social sciences generality may be limited by social conditions.
- **Predictability:** Scientific knowledge should help predict outcomes when similar conditions exist. Prediction depends on stable cause-and-effect relationships.
- **Objectivity:** Results should not be influenced by the researcher's personal opinion, belief or bias. Different observers should reach similar conclusions from the same evidence.
- **Systematic procedure:** Scientific inquiry follows an accepted, planned sequence of investigation. Haphazard or accidental discovery cannot be called fully scientific.
- **Empirical basis:** Scientific conclusions are based on facts, data, observation and evidence rather than tradition or speculation.
- **Logical reasoning:** Scientific method uses induction and deduction to build, test and refine knowledge.

  Exam point: In exam answers, remember the five core terms: verifiability, generality, predictability, objectivity and system.

1.3 Differentiate between research method and research methodology.

Research methods are the specific tools and techniques used to conduct research, while research methodology is the broader logic and scientific explanation behind the selection and use of those methods.

Basis	Research Method	Research Methodology
Meaning	Specific techniques used in research operations.	Scientific study of how research is done and why particular methods are chosen.
Scope	Narrower in scope.	Wider in scope. It includes methods plus reasoning, assumptions and justification.
Focus	Focuses on how data are collected, measured and analyzed.	Focuses on why a particular approach, design, tool or analysis is appropriate.
Examples	Questionnaire, interview, observation, sampling, t-test, chi-square test.	Explanation of why a questionnaire was used, why a sample design was selected, and why a test is suitable.
Role	Helps to obtain data and solve the immediate research problem.	Helps to design a scientifically valid and defensible research process.

Thus, methods are the instruments of research, whereas methodology is the science and logic of using those instruments correctly.

1.4 Differentiate between fundamental/basic research and applied research with suitable examples.

Research may be classified on the basis of purpose into fundamental/basic research and applied research.

Basis	Fundamental / Basic Research	Applied Research
Purpose	To add new knowledge, develop theory and increase understanding.	To solve a specific practical problem.
Use of findings	Findings may not have immediate practical application.	Findings are directly useful for decision-making and action.
Nature	Theoretical and generalizable.	Practical, problem-oriented and action-oriented.
Time focus	Long-term contribution to knowledge.	Immediate or short-term solution.
Example	Studying why employee commitment differs across age groups.	Finding the best strategy for a company to improve productivity.

Applied research uses scientific methods to solve real-life problems, while basic research improves the body of knowledge that may later support applied work.

1.5 Explain the difference between concept, construct and variable.

Concepts, constructs and variables are important parts of research language.

- **Concept:** A general idea or symbol representing an object, event, condition or phenomenon. Examples: democracy, leadership, productivity, job satisfaction.
- **Construct:** A higher-level abstract concept created for a specific research purpose. Constructs cannot be observed directly; they must be inferred from observable indicators. Examples: intelligence, motivation, organizational commitment.
- **Variable:** A measurable characteristic that can take different values across persons, groups or situations. Examples: age, income, level of education, job satisfaction score.

Basis	Concept	Construct	Variable
Level of abstraction	General idea.	More abstract and theoretical.	Operational and measurable.
Observation	May be concrete or abstract.	Usually cannot be directly observed.	Can be measured or categorized.
Use in research	Gives meaning to a phenomenon.	Provides theoretical explanation.	Used in hypotheses, data collection and analysis.
Example	Job satisfaction.	Work motivation.	Pay, promotion score, satisfaction score.

A concept becomes useful in research when it is converted into measurable variables through operationalization.

Unit 2: Research Problem Identification and Formulation

2.1 Define research problem. Explain the steps followed in identifying a research problem.

A research problem is a clear, specific and researchable issue, difficulty or question that a researcher wants to investigate scientifically. It indicates what is to be studied and gives direction to the whole research process.

Steps followed in identifying a research problem are:

9. **Determine the field of research:** Select the broad area in which the researcher is interested.
10. **Develop mastery in the area:** Gain basic knowledge of theories, concepts and current issues in the selected field.
11. **Review recent research:** Study journals, reports, theses and other literature to know what has already been done.
12. **Select a priority area:** Choose an area where there is a gap, contradiction, practical problem or need for further study.
13. **Use discussion and experience:** Discuss with supervisors, experts and colleagues, and use personal field experience to locate a meaningful issue.
14. **Pinpoint the specific aspect:** Narrow the broad topic into a precise, manageable and testable research problem.

A research problem may arise from personal experience, literature review, social changes, technological innovation, gaps in previous studies or suggestions given in earlier research reports.

2.2 What are the criteria of a good research problem/problem statement?

A good research problem should be clear, meaningful and capable of scientific investigation. The major criteria are:

- **Relation between variables:** The problem should express or imply a relationship between two or more variables.
- **Clarity and precision:** The problem should be stated clearly, rigidly and unambiguously so that its meaning is not confusing.
- **Empirical testability:** It should be possible to collect data and test the problem through observation, measurement or analysis.
- **Feasibility:** The problem should be manageable within available time, money, skill, data and resources.
- **Significance:** The problem should contribute to knowledge, practice, policy or decision-making.
- **Originality:** It should not simply repeat a completed study unless there is a new context, method or gap.
- **Link with research design:** The problem should suggest the appropriate design, data and method of analysis.

A good problem statement usually raises a question about the relationship between variables, states the relationship clearly and suggests a method of researching the question.

2.3 Define hypothesis. Explain different types of hypothesis.

A hypothesis is a tentative, testable and logical statement that predicts the relationship between two or more variables. It is a working assumption that guides the researcher about what data to collect and what relationship to test.

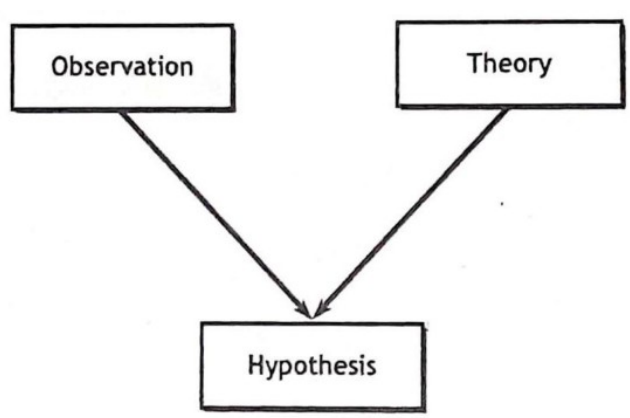

Figure: Hypothesis can be developed from both observation and theory (from the study material, page 20).

Important types of hypothesis are:

- **Descriptive hypothesis:** States the existence, size or distribution of one variable. Example: Tribhuvan University is experiencing budget difficulties.

- **Relational hypothesis:** States a relationship between two or more variables. Example: Higher income families spend more on recreation.
- **Correlational hypothesis:** States that variables are associated but does not necessarily prove cause and effect. Example: Education is positively related to political participation.
- **Explanatory/causal hypothesis:** States that one variable causes or influences another. Example: Increased workload reduces job satisfaction.
- **Directional hypothesis:** Specifies the direction of relationship or difference. Example: Younger workers are less motivated than older workers.
- **Non-directional hypothesis:** States that a relationship or difference exists but does not specify direction. Example: There is a difference in work attitudes of industrial and agricultural workers.
- **Null hypothesis:** States no relationship, no difference or no effect. It is tested statistically.
- **Alternative hypothesis:** States that a relationship, difference or effect exists. It is supported if the null hypothesis is rejected.

  Exam point: A good hypothesis should be declarative, clear, testable, limited in scope and based on theory, previous research or logical reasoning.

2.4 Differentiate between research hypothesis and statistical hypothesis.

A research hypothesis and a statistical hypothesis are closely related, but they are not the same.

Basis	**Research Hypothesis**	**Statistical Hypothesis**
Meaning	A substantive statement predicting a relationship between variables in the research problem.	A formal statement about a population parameter that can be tested using statistical techniques.
Language	Expressed in conceptual/research language.	Expressed in statistical symbols or parameter terms.
Example	Female students have higher academic performance than male students.	H0: mu1 = mu2; H1: mu1 > mu2.
Function	Guides the study and data collection.	Guides statistical testing and decision-making.
Basis	Derived from research question, theory or literature.	Derived from the research hypothesis for statistical analysis.

In practice, the research hypothesis is converted into null and alternative statistical hypotheses before testing.

2.5 Explain null hypothesis and alternative hypothesis with examples.

A null hypothesis, usually written as H0, is a statistical statement that there is no significant difference, no relationship or no effect. It is assumed true until statistical evidence suggests otherwise. An alternative hypothesis, written as H1 or HA, is the opposite of the null hypothesis and states that a difference, relationship or effect exists.

- **Example 1:** H0: There is no relationship between pay and productivity. HA: Pay and productivity are positively related.
- **Example 2:** H0: There is no difference between male and female workers in productivity. HA: Male and female workers differ in productivity.
- **Example 3:** H0: Working conditions have no influence on job satisfaction. HA: Better working conditions improve job satisfaction.

The function of the null hypothesis is to provide a neutral statement for statistical testing. Researchers usually try to reject H0 in order to support HA. If evidence is insufficient, H0 is not rejected.

  Exam point: Null hypothesis represents the no-effect position; alternative hypothesis represents the expected research claim.

Unit 3: Review of Literature and Research Design

3.1 Define literature review/literature survey. Explain its purposes, benefits and functions.

A literature review is a systematic and critical study of published books, journal articles, reports, theses and other sources related to a research problem. It summarizes, compares, evaluates and synthesizes previous knowledge in order to locate the present study within an existing field.

Purposes and benefits of literature review are:

- To know what research has already been done on the topic.
- To understand important theories, concepts and debates in the field.
- To identify gaps, contradictions and unanswered questions.
- To avoid unnecessary duplication of completed research.
- To identify variables, data sources, researchable hypotheses and suitable methods.
- To develop a theoretical framework for the study.
- To justify the need and importance of the proposed research.
- To improve research design by learning from strengths and weaknesses of previous studies.

Functions of literature review include:

- Ensures that the researcher is not reinventing the wheel.
- Gives credit to scholars who have laid the foundation for the study.
- Demonstrates the researcher's knowledge of the problem area.
- Shows the ability to critically evaluate and synthesize existing literature.
- Helps convince readers that the proposed research will make a meaningful contribution.

  Exam point: A strong literature review is critical, organized and analytical; a weak literature review is only a list or summary of sources.

3.2 What is research design? Explain the essential elements of research design.

Research design is the overall plan, structure and strategy of investigation prepared to answer research questions or test hypotheses. It is the blueprint for collecting, measuring and analyzing data.

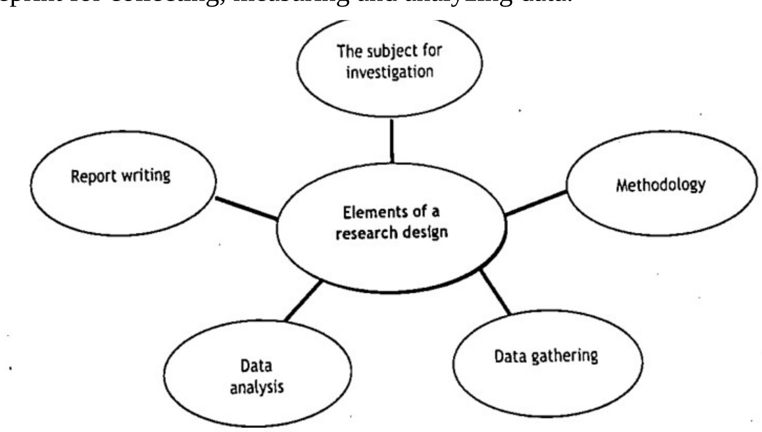

Figure: Basic elements of research design (from the study material, page 35).

Essential elements of research design are:

- **Problem/subject for investigation:** The design begins with a clear research problem or question.
- **Methodology:** The researcher decides the approach, design type, methods, tools and procedures.
- **Data gathering:** The design specifies what data are needed, where they will be obtained, from whom and through which instruments.
- **Data analysis:** It states how data will be coded, classified, tabulated and analyzed.
- **Report writing:** It includes how findings will be organized, interpreted and communicated.

A good research design also includes sampling plan, measurement of variables, reliability and validity of tools, data collection schedule, ethical issues and statistical techniques.

3.3 Explain the steps involved in the research design process.

The research design process converts a research idea into a workable plan. The major steps are:

15. **Define the research problem:** Clearly state what is to be investigated.
16. **Review literature:** Study previous research to identify gaps, variables and methods.
17. **Formulate objectives, questions and hypotheses:** Specify what the study will answer or test.
18. **Choose the research approach/design:** Select exploratory, descriptive, analytical, experimental, survey, case study or other suitable design.
19. **Define population and sample:** Identify target population, sampling frame, sampling technique and sample size.
20. **Select data collection tools:** Choose questionnaire, interview schedule, observation, records or measurement scales.
21. **Plan measurement of variables:** Operationally define variables and decide scales of measurement.
22. **Plan data collection procedure:** Decide where, when, how and by whom data will be collected.
23. **Choose data analysis technique:** Select appropriate statistical tests, models or qualitative analysis method.
24. **Plan interpretation and reporting:** Decide how findings will be interpreted in relation to the research problem.

A research design must be feasible, economical, valid and suitable for the objectives of the study.

3.4 What is case study research? Describe its characteristics and limitations.

Case study research is an in-depth investigation of a particular individual, group, institution, event, community or phenomenon within its real-life context. It studies a social unit as a whole and uses multiple sources of evidence such as interviews, documents, observation and records.

Characteristics of case study research are:

- It is intensive and detailed.
- It studies one or a few units deeply rather than many units superficially.
- It examines the present status, past background and environmental influences of the case.
- It uses multiple sources of data.
- It helps identify important variables, processes and interactions.
- It is useful for exploratory research and for developing hypotheses for further study.
- It studies the phenomenon in its natural context.

Limitations of case study research are:

- Findings from a single case cannot be generalized to the whole population without follow-up research.
- It may be expensive and time-consuming.
- It may involve subjectivity or researcher bias.
- The narrow focus may miss broader patterns.
- It depends heavily on the skill and judgment of the researcher.

  Exam point: Case study gives depth, not wide generalization.

3.5 Explain experimental research method and mention different types/steps of experimental research.

Experimental research is a method used to study cause-and-effect relationships. The researcher manipulates one or more independent variables and observes their effect on the dependent variable while controlling other factors as far as possible.

In experimental research, the group receiving the treatment is the experimental group and the group used for comparison is the control group. It is especially useful when the aim is to test whether changes in variable A cause changes in variable B.

Types of experiments include:

- **Positive and negative experiment:** A positive experiment has both phenomenon and cause present; a negative experiment removes the cause to observe whether the phenomenon disappears.
- **Natural experiment:** Conducted in natural conditions where the researcher observes naturally occurring differences or changes.

- **Laboratory experiment:** Conducted in an artificial, controlled environment where variables can be manipulated precisely.
- **Field experiment:** Conducted in a natural setting, but with some manipulation and control by the researcher.

Steps in experimental research are:

25. State the problem, research questions and objectives.
26. Review literature and identify possible outcomes and variables.
27. Formulate hypothesis.
28. Design the experiment, including treatment, control group, sample and measurement.
29. Perform the experiment using control, randomization, replication and blocking where possible.
30. Analyze experimental outcomes statistically.
31. Draw conclusions by testing significance and reliability.
32. Check validity of results and compare with standards or similar studies.
33. Evaluate the investigation through practice or repetition.

Ethical issues must be considered, such as voluntary participation, informed consent, right to withdraw, no physical or mental harm, confidentiality and proper debriefing.

Unit 4: Data Analysis and Interpretations

4.1 Explain nominal, ordinal, interval and ratio scales with suitable examples.

Measurement is the assignment of numerals or symbols to objects, events or responses according to rules. The four major levels of measurement are nominal, ordinal, interval and ratio scales.

Scale	Meaning	Example	Main statistical use
Nominal	Classifies data into mutually exclusive categories without rank.	Gender: male/female; religion; department.	Frequency, percentage, mode, chi-square.
Ordinal	Ranks objects or responses in order, but exact difference between ranks is not known.	Rank preferences for cities, social status ranking, Likert categories.	Median, percentile, rank correlation, non-parametric tests.
Interval	Has ordered categories with equal intervals, but no true zero point.	Temperature in Celsius; attitude or intelligence scores.	Mean, standard deviation, correlation, t-test, F-test.
Ratio	Has all properties of interval scale plus an absolute zero.	Age, income, number of children, weight, time, sales.	All mathematical and statistical operations including ratio comparison.

The level of measurement determines what statistical techniques can be used. Nominal is the simplest scale, while ratio is the most powerful scale.

4.2 What is ratio scale? Explain its superiority over other scales of measurement.

A ratio scale is the highest and most powerful level of measurement. It has all the properties of nominal, ordinal and interval scales, and it also has an absolute or true zero. Because of true zero, meaningful ratio comparisons can be made.

Examples of ratio scale are age, income, number of workers, number of children, weight, height, length, time, distance and production quantity.

Superiority of ratio scale:

- It classifies objects into categories like nominal scale.
- It ranks values from lower to higher like ordinal scale.
- It has equal intervals like interval scale.
- It has a true zero point, which interval scale lacks.
- It allows meaningful statements such as “twice as much” or “half as much”.
- It permits all arithmetic operations: addition, subtraction, multiplication and division.
- It allows the use of advanced statistical measures such as mean, standard deviation, coefficient of variation, geometric mean and harmonic mean.

For example, a person earning Rs. 80,000 has twice the income of a person earning Rs. 40,000. Such comparison is possible only in ratio scale because zero income has real meaning.

4.3 Define reliability. Describe the various types/methods of reliability.

Reliability means consistency, stability and dependability of a measuring instrument. A scale or test is reliable if it gives the same or similar results when used repeatedly under similar conditions.

Methods of estimating reliability are:

- **Test-retest method:** The same test is administered to the same group at two different times. The correlation between the two sets of scores indicates stability over time.
- **Parallel/alternate forms method:** Two equivalent forms of the same test are given to the same group. A high correlation between the two forms indicates reliability.
- **Split-half method:** The test is divided into two equal halves, such as odd and even items. Correlation between the two halves is calculated and adjusted using the Spearman-Brown formula.

- **Rational equivalence / Kuder-Richardson method:** Reliability is estimated through internal consistency among items in the same test, especially for items scored right/wrong.
- **Internal consistency:** Checks whether different items in a scale measure the same construct. Cronbach's alpha is commonly used for this purpose in modern research.

A reliable instrument reduces random measurement error. However, reliability alone does not guarantee validity.

> **Exam point:** Reliability answers the question: Does the instrument measure consistently?

4.4 Define validity. Describe the various types of validity.

Validity means the degree to which a measuring instrument actually measures what it is intended to measure. A scale is valid if its scores reflect the true differences in the characteristic being measured.

Major types of validity are:

- **Content validity:** The test items adequately represent the whole content or domain being measured. For example, an exam in research methodology should cover all important units of the syllabus.
- **Criterion-related validity:** The measure is compared with an external criterion known to be valid. It has two forms: predictive validity and concurrent validity.
- **Predictive validity:** The ability of a test to predict future performance or outcome. Example: an aptitude test predicting future academic performance.
- **Concurrent validity:** The degree to which a new measure correlates with an already validated measure at the same time.
- **Construct validity:** The degree to which a test truly measures the theoretical construct it claims to measure, such as intelligence, motivation or job satisfaction.

Reliability and validity are related. A valid measure must be reliable, but a reliable measure may not be valid. For example, a weighing machine may give the same wrong weight every time; it is reliable but not valid.

> **Exam point:** Validity answers the question: Does the instrument measure the right thing?

4.5 Distinguish between sampling error and non-sampling error. How can they be reduced?

Errors in research data may be classified into sampling errors and non-sampling errors.

Basis	Sampling Error	Non-sampling Error
Meaning	Error that occurs because only a sample, not the whole population, is studied.	Error that occurs during planning, data collection, response, recording, processing or reporting.
Presence	Present only in sample surveys.	Present in both census and sample surveys.
Cause	Chance variation between sample statistic and population parameter.	Wrong questionnaire, response bias, interviewer bias, non-response, coverage error, coding and tabulation mistakes.
Measurement	Can often be estimated statistically as standard error or margin of error.	Difficult to measure exactly.
Effect of larger sample	Usually decreases when sample size increases.	May not decrease with larger sample; may even increase if fieldwork is poorly controlled.

Ways to reduce sampling error:

- Use probability sampling wherever possible.
- Increase sample size when greater precision is required.
- Use stratified sampling when the population is heterogeneous.
- Use a complete and updated sampling frame.
- Avoid faulty selection and substitution of sample units.

Ways to reduce non-sampling error:

- Define objectives and population clearly.
- Prepare clear, brief and unbiased questionnaires.

- Train investigators and supervise fieldwork.
- Pretest the questionnaire before final survey.
- Reduce non-response through follow-up and proper contact methods.
- Check data through editing, coding verification, consistency checks and careful tabulation.
- Maintain confidentiality to reduce false responses and prestige bias.

Unit 5: Preparation of Research Report

5.1 What is a research report? What purposes does it serve? Why is reporting of research study important?

A research report is a clear, concise and organized written document that communicates the process, data, analysis, findings and conclusions of a research study. It is the final tangible product of research work.

Purposes of research report are:

- To communicate research findings to a specific audience.
- To present data, analysis and conclusions in an organized form.
- To provide evidence for academic evaluation, policy-making or managerial decision-making.
- To allow other researchers to understand, verify, replicate or extend the study.
- To contribute to the existing body of knowledge.
- To preserve the research process and findings for future reference.

Importance of reporting research:

- Research has little value if its findings are not communicated properly.
- The quality of the research is often judged by the quality of the report.
- A good report helps readers judge the adequacy of methods and seriousness of findings.
- It shows the researcher’s ability to organize ideas, analyze evidence and write objectively.
- It helps transform raw data into useful knowledge.

A good research report should be accurate, objective, complete, well-structured, clear and written according to academic conventions.

5.2 Explain the goal and structure of a research proposal.

A research proposal is a written plan of a research project that explains what the researcher intends to study, why the study is important and how the study will be conducted. Its goal is to convince a supervisor, department, funding agency or academic body that the proposed research is relevant, feasible and worthwhile.

Goals of a research proposal include:

- To present the research problem and justify its importance.
- To show that the researcher understands the field and literature.
- To identify the gap the study will address.
- To explain the objectives, questions and hypotheses.
- To describe the research design, methods, sampling and data analysis plan.
- To show feasibility in terms of time, resources and skills.
- To get approval or funding for the study.

Common structure of a research proposal:

34. Title page
35. Abstract or summary, if required
36. Introduction and background
37. Statement of the problem
38. Research objectives and research questions
39. Hypotheses, if applicable
40. Review of literature
41. Significance of the study
42. Research design and methodology
43. Population, sample and sampling method
44. Tools and techniques of data collection
45. Data analysis plan
46. Work schedule or timeline
47. Expected contribution or implications
48. References or bibliography

49. Appendices, if needed

5.3 Explain the contents/steps of a research proposal with brief explanation.

The contents of a research proposal may vary by institution, but the essential sections are usually similar.

- **Title:** Gives a clear and concise idea of the research topic.
- **Introduction:** Introduces the topic, context and background of the study.
- **Statement of the problem:** Explains the exact issue or gap the research will address.
- **Research objectives:** States the general and specific aims of the study.
- **Research questions/hypotheses:** Shows what the study seeks to answer or test.
- **Review of literature:** Summarizes and evaluates previous research related to the topic.
- **Significance of study:** Explains the academic, practical, social or policy importance of the research.
- **Research methodology:** Describes design, population, sample, sampling technique, data collection tools and procedures.
- **Data analysis plan:** Explains the statistical or qualitative methods to be used for analysis.
- **Scope and limitations:** Clarifies boundaries of the study and possible constraints.
- **Time schedule:** Shows the planned sequence and duration of research activities.
- **References:** Lists the sources cited in the proposal using a required style.
- **Appendices:** Includes tools such as questionnaire, interview schedule or additional documents.

A good proposal should be brief, clear, focused, well-cited and logically connected from problem to method.

5.4 What are the various methods of documenting sources? Describe APA method of citation with examples.

Documenting sources means acknowledging the books, articles, reports, websites and other materials used in research writing. It prevents plagiarism and allows readers to locate the original sources.

Common methods or styles of documenting sources include:

- APA style - widely used in social sciences and education.
- IEEE style - uses numbered citations in square brackets, common in engineering and technology.
- MLA style - widely used in humanities.
- Chicago/Turabian style - used in history and some social sciences.
- Harvard style - author-date referencing style.

APA citation uses the author-date system in the text and a detailed reference list at the end. In-text citation includes the author's surname and year of publication.

- **In-text citation - paraphrase:** Kumar (2011) explains that research methodology guides the systematic solution of a research problem.
- **In-text citation - parenthetical:** Research design is the plan and procedure for conducting research (Creswell, 2014).
- **Book reference format:** Author, A. A. (Year). Title of the book. Publisher.
- **Book example:** Kothari, C. R. (2014). Research methodology: Methods and techniques. New Age International Publishers.
- **Journal article format:** Author, A. A., & Author, B. B. (Year). Title of article. Title of Journal, volume(issue), page range.
- **Journal article example:** Shrestha, P., & Rai, M. (2020). Sampling practices in social research. Journal of Research Studies, 12(2), 45-58.

In APA style, the reference list is normally arranged alphabetically by the surname of the first author.

5.5 Explain plagiarism and ethical issues in research and publishing articles.

Plagiarism is the act of using another person's words, ideas, data, structure or work without proper acknowledgement. It includes copying text, paraphrasing without citation, using someone else's research design or data without permission, and submitting another person's work as one's own.

Types or forms of plagiarism include:

- Direct plagiarism - copying exact words without quotation and citation.

- Paraphrasing plagiarism - changing a few words but keeping the same idea without credit.
- Self-plagiarism - reusing one's own previous work without disclosure or citation.
- Mosaic plagiarism - mixing copied phrases with original writing.
- Data plagiarism - using another researcher's data, tables or figures without permission or citation.

Ethical issues in research include:

- Obtaining informed consent from participants.
- Protecting privacy and confidentiality of respondents.
- Avoiding physical, psychological or social harm.
- Allowing participants to withdraw from the study.
- Avoiding deception unless justified and ethically approved.
- Using data honestly and not fabricating or falsifying results.
- Giving proper credit to authors, institutions and funding bodies.
- Reporting findings objectively, including limitations.
- Avoiding duplicate publication and salami publication.
- Declaring conflicts of interest.

Ethical issues in publishing articles include proper authorship, avoiding plagiarism, accurate citation, honest peer-review conduct, no data manipulation, no duplicate submission to multiple journals and correction of errors after publication.

> **Exam point:** Plagiarism can be avoided by careful note-taking, quotation marks for exact words, paraphrasing properly, citing all sources and using reference-management tools.

Quick Revision Matrix

Unit	Highest-yield answer themes
Unit 1	Scientific research process; scientific method; method vs methodology; applied vs basic; concept/construct/variable.
Unit 2	Research problem; criteria of good problem; hypothesis types; research vs statistical hypothesis; null vs alternative.
Unit 3	Literature review; research design; case study; experimental research; citation and referencing.
Unit 4	Measurement scales; ratio scale; reliability; validity; sampling and non-sampling errors.
Unit 5	Research report; research proposal; proposal contents; APA citation; plagiarism and ethics.

Use these answers as a base. In the exam, start each answer with a definition, then write points with short explanations and examples. For long answers, end with a one-line conclusion.